

Project Title and Description

The name of the term project is “Smash112” which is a platform fighting game. This is a multiplayer game that enables two players to control characters in the game to fight against each other. The game will begin with a page that has instructions telling players the basic game rules: how to choose character, how to choose a fighting field, and keys control the character. Then, the player will enter a page that enables them to choose the character and fighting field. In this process, players can also press a designated key to view the special abilities each character has.

When players proceed to the actual fighting, there will be a health bar on each side of the laptop screen telling them how many life points their character has. Players can use the attacking ability and the defending ability of their characters to knock out the other side or defend the attack from the other player. When one player’s health bar first reaches zero. The game is over and the word game over will show up on the screen. Then, the game will enter a summary page that honors the winner. After this page, the players will have the choice to restart the whole game or have another game controlling the same character and fighting in the same field.

Moreover, AI players are also added to the game. Using the behavior tree algorithm, the AI player will control the other character to play against the human player. The AI player will also have different difficulty levels by prioritizing different values in the behavior tree.

Similar Project:

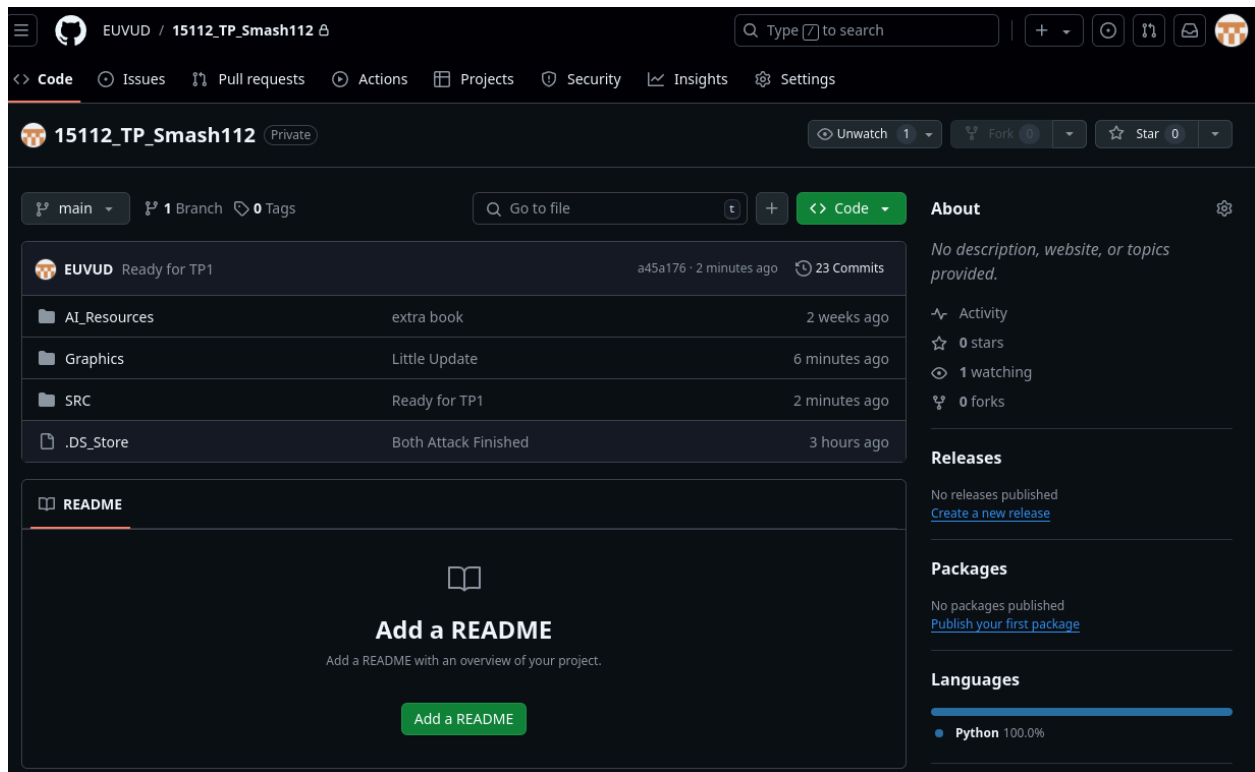
SmashBro, Stickman Fighter: Epic battle, and Guilty Gear are studied to inspire the design of the game. The abilities of different characters in the game are studied and possibly recreated on the characters in my own game. For example, in Smash Bro every character has some special abilities to attack other characters. There is a character named “Kirby”. It has the ability to copy other characters’ abilities and gain their special moves. There is also another character named “Link” who can throw remote bombs at the enemy. I also studied many other characters’ abilities in SmashBro and will incorporate them into my own game.

The fighting field can also be recreated in my game. In Stickman Fighter, it has a very complicated fighting field. There are spikes on the ground that can deal damage to the player. There is also a fighting field completely made up of moving blocks which require players to carefully control the character from falling off. Moreover, sometimes, there will be random weapons falling from the sky; players can control their characters to pick up these weapons to deal more damage on their opponents.

I also studied the character and field selection page’s design in these games and will design a new selection page of my own. SmashBro has a very simple and beautiful field selection page where players can move the mouse on a map of different fighting fields. When the mouse is on the specific fighting field, there will be an enlarged view on the left side of the screen. I will try to

remake this field selection feature into my own game. I will design a character manual page where players can see the specific ability each character has and how to control characters. I will also study the keys guilty gear used to control the character. For example, 'WASD' will be used as the movement since these are the conventional keys. Number keys such as 1 and 2 will be used for punch and kick since it prevents two players from clicking on keys that are too close. All these keyboard controls will be incorporated into my game so my players can have an easier time adapting to my key control design to play with my characters.

Version Control:



All versions of my game are backed up daily into my private Github repository. I normally back up every time there is a major function or change added to my game. And I will try to maintain a daily update if there is any change made on my game.

Tech List:

No external modules/hardware/tech will be used in this process.

Storyboard

Start Page

Smash-112

Press Any Button to Start

Game Mode Selection

AI Mode

Multi-Player Mode

Fighting Field Selection

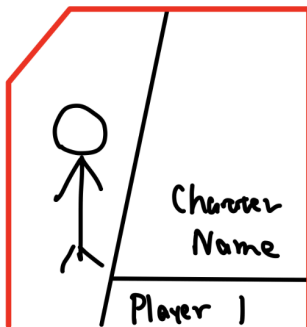
Field Name

Field Selection

Selected Field	
	?

Character Selection

Characters			
	 P_1		
		 P_2	?



Fighting Field



Pause Page

Resume

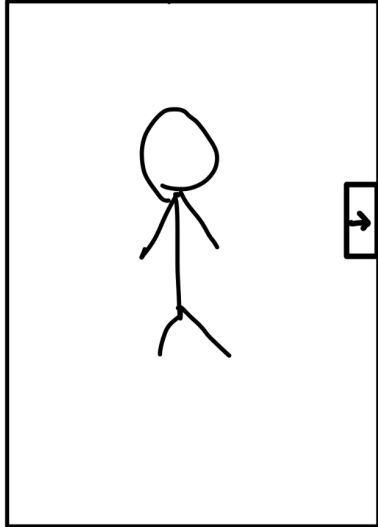
Settings

Quit



User Manual

Character



Basic Instruction

Ability 1

Ability 2